

ANIMA

A Developmental Approach to a Self-Organizing Forager

Technical white paper · Research record · Reproducible

ANIMA: A Developmental Approach to a Self-Organizing Forager

Experimental Findings, Registered Negative Results, and the Two Walls

Status: White paper · Research record · September 2026 **System:** `world\_survival` (crates/anima-world), ANIMA developmental artificial nervous system in Rust **Reproducibility:** deterministic (seeded), byte-identical defaults, all mechanisms env-gated

1. Executive Summary

ANIMA is a from-scratch, self-organizing artificial nervous system: a spiking LIF network receives retinal/proprioceptive frames from a physical 3D body in an arena and controls it to forage for food. Under a binding charter it uses **no hand-designed internal architecture, no pretrained models, no external memory modules**, and the only reward is food.

This paper records a full characterization campaign. We found and fixed a real control bug, built a working forager, gave it a second drive (a home/rest site), and confirmed it can produce **repeated return-home behavior when that is rewarded**. But we also isolated — with measured evidence, not inference — **two walls** that block the original open-ended goal:

1. **Internal-band neural saturation**: the network's internal (expansion) layer fires at 100% every tick (`out\_sum 6000`, `kc\_spikers 256/256`). Every reward-gated learning mechanism therefore becomes a non-selective global wash that the evolutionary loop ignores. 2. **Fitness-economy ceiling**: evolution pins at ~121-137.6 fitness regardless of mechanism stack.

A four-lever campaign (competition, spike threshold, input amplitude, per-neuron heterogeneity) failed to de-saturate the band — and revealed it is **temporally bistable**: cold→sparse, then a sharp phase-transition to hot→saturated. Selective learning requires sustained sparsity and is therefore structurally absent. Tracked next levers are listed in §9.

2. Motivation

Goal: an organism that autonomously forages — finds food, consumes it — and, ultimately, self-improves open-endedly ("make it like us"). The project favors **laboratory discipline over best-effort results**: every mechanism is registered, falsifiable, and reproducible.

3. The Agent and the World

- Body**: 6-DOF physics (yaw/pitch/thrust/strafe/lift/brake), arena 100×100, gravity 9.81 applied in the simulator only, ground clamp, energy tank (starts 100, drain 1.0/beat, death at 0).
- Sensory**: 48-channel retina (5 primitives, analytic ray raster) + 8 proprioceptive channels, plus optional 6-channel food-smell (body-relative gradient).
- Network**: LIF spiking, 1 ms ticks, seeded `Xoshiro256PlusPlus`, STDP + structural growth (V2), 24-56 inputs / 40-(128)-(256) internal / 12 motor outputs.

- **Reward**: food only. Steering is the action, never rewarded directly.
- **World dynamics**: food placement (static seed / curriculum / escalating / stable), energy refill on contact, death at tank zero.

4. Binding Constraints (the Charter)

- No hand-designed internal architecture; no pretrained/CNN/CLIP; no external memory modules.
- All mechanisms **env-gated**; with every gate unset, output is byte-identical to a frozen baseline.
- Determinism: same args + env $\Rightarrow$ identical output (registered).
- **Pre-registration**: expected direction and success bars are set before running; negatives are recorded, constants are not tuned on outcomes.
- Charter amendments (e.g. adding a home need) require explicit operator approval.

5. Method

Protocol. Each mechanism is introduced env-gated, then verified: (1) default identity, (2) determinism (two-run diff), (3) observability (gate on vs off), (4) workspace suite, (5) a registered success bar. Failures are recorded as evidence-backed negatives — never silently retuned. Results below are measured; unverified causal claims are flagged as such.

6. Measured Results

6.1 Verified wins

#	Mechanism	Result	Evidence
1	Control bug fix	Steering yaw channel was sign-inverted; body flew away from food	post-fix food_dist 11.5 $\rightarrow$ 6.7, touches 0 $\rightarrow$ 9 at reachable radius
2	Foraging under shaping	Up to 126 touches / life; strong local pursuit	curriculum + smell stack
3	SL2_STASIS (stable food + re-entry guard + tank cap)	3 $\rightarrow$ 36 genuine touches / life; energy bounded; no farm	unsaturable only via geometry-exact re-entry guard
4	SL2_HMEXP (hunger-modulated search)	settle-when-fed (0.4 $\times$ noise floor), roam-when-hungry; early-survivor signal in	single-life $\approx$ baseline; evolve best 0 $\rightarrow$ 33.9 by gen 5 (8-gen window)
5	SL2_NEST + SL2_NESTFIT (home/rest second drive + homing)	First repeated return-home behavior (elite ret 1 $\rightarrow$ 3 on seed 7)	home = nearest non-food landmark; stamina = capability constraint; reward per return
6	Proactive survival (SL2_PROG)	survived 2230 beats (vs $\sim$ 60–200), food escalates until defeated	big lifespan + escalating difficulty

6.2 Registered negative results

Mechanism	Outcome	Notes
SL2_NOVELTY (MCC archive)	observably steers selection; ceiling unchanged	rank-order changes, no escape
SL2_RNDPP (dopamine $\times$ STDP)	fires correctly (4.9 $\times$ Δw in window) but invisible to	$\sim$ 1e-3/synapse below mutation floor
SL2_MARS / SL2_DENSITY	environmentally observable; ceiling unchanged	same survivor, bigger number
SL2_MB (mushroom-body locus)	wiring verified firing; no sparse premise	256/256 KCs eligible — non-selective
De-saturation campaign (§7)	all four levers bounded negative	competition / threshold / amplitude / heterogeneity

Cross-cutting finding: within-life plasticity (any locus) is $\sim$ 1e-3–0.24/synapse but the GA's selection is **death-gated and coarse** ($\pm$ 10% mutation) — so learning that is not selectivity-visible never changes outcomes. Evolution "listens" only to selection-level changes (novelty rank-order, the homing reward NESTFIT — the one lever that moved behavior).

7. The Two Walls

7.1 Wall 1 — Internal-band saturation $\Rightarrow$ selective learning is impossible

The internal/expansion layer fires at 100% every tick. Under this regime any reward-gated plasticity is a **uniform global wash** the GA trivially mutates around (measured: evolve trajectories byte-identical with and without the reward bump).

De-saturation campaign (all measured):

Lever	Range	Result	Mechanism
Pairwise competition (SL2_INH/SL2_INH)	0.5–4.0	256/256 always	Poisson drive rebuilds the one-tick inhibitory dip
Spike threshold (SL2_THETA)	2–8×	256/256 always	52-amplitude input drive crosses any spread
Input amplitude (SL2_AMPL)	52→3	100%→0% cliff	uniform scaling is bistable — no graded regime
Heterogeneity (SL2_HET)	0.2–1.0	256/256 alone; transient at corner	lognormal threshold spread can't sustain

Final mechanism (discriminator, KCPROBE EVERY=10): the band is **temporally bistable** — cold at life start (2/256), then a **sharp phase-transition to saturated** by beat ~25–30. It is a collective recurrent-excitation/hysteresis flip, not a gradual slow-state ramp (a linear rate-constant knob likely won't tame it).

7.2 Wall 2 — Fitness-economy ceiling

Evolution converges to a scoring-family-dependent pin (not one flat number):

- Plain-survival scoring** (``final_energy + 3·touches``): pins 121.8 / 121.0 (HMEXP+NEST, seeds 1/2).
- Density/ARS scoring** (adds ``20·touches²/beats`` and ``4·persistence``): pins 137.5–137.6

(ARS run seed 7; the higher number is the bonus terms, not a real escape).

- NESTFIT scoring** (plain + ``40·nest_returns``): 121.8 / 179.4 (seed 1 / seed 7), where 179.4

includes ``+40×3`` from three returns — again additive, not a genuine survival gain.

``final_energy + 3·touches`` saturates once food is reachable; across every scoring family the ceiling is the fitness-economy saturation, and the ~150–180 bests are additive bonuses.

8. Discussion

- The organism is **real and capable**: it forages strongly, survives long, and can sustain a home-return behavior *when rewarded* — the first causal win for the "second drive" direction.
- The walls are **structural, not tuning**: saturation is immune to competition/threshold/amplitude/heterogeneity at every tested magnitude, and the fitness economy is a property of the reward shape.
- The iron rule of the campaign: **within-life learning cannot clear a coarse, death-gated selection**; only selection-level changes are evolution-visible.
- Full-stack non-additivity**: running the coherent verified organism

(``STASIS` + `HMEXP` + `NEST` + `NESTFIT` + `OINH` + `pool128``) reproduces the NESTFIT baseline exactly at 15 gens (seed 20260924: 121.8/ret=0; seed 7: 155.0/ret=1) — the mechanisms are orthogonal to the fitness economy, not additive.

8.5 Limitations

The claims above are bounded by the measurement window and should be read with these limits:

- Small N seeds**: only two evolve seeds per run; seed-dependence (e.g. ``elite_ret` 0 vs 3`) is a real finding but a 2-seed sample cannot establish robustness.
- Short generation windows**: several series (MB, full stack) ran 12–15 gens per seed within a 3500 s tool cap; pin values at longer horizons (e.g. 179.4/ret=3 at gen 20–30) come from a subset of runs.
- Single-threaded CPU stack**: 256-pool evolution exceeded the hour cap; some negatives (MB locus, larger expansions) were bounded by compute, not measured to conclusion.

- **Partial windows**: a few registered negatives rest on overlapping-generation comparisons (e.g. gens 0-5) rather than full-architecture sweeps.
- No GPU, no parallelization; determinism is enforced serially.

9. Recommended Next Steps (registered levers)

1. **Attack the phase-transition, or keep the band cold**: target the recurrent-excitation/hysteresis that flips the band to saturation, or maintain continuous-cold sparsity — the precondition for any selective-learning re-test (``SL2_MB`` locus remains unproven, not falsified). 2. **Reform the fitness economy**: sustained-foraging/ survival density shaping so selection has a gradient to climb past the ceiling. 3. **GPU/parallel step** (engineering): the fly-connectome literature runs 138k neurons real-time on GPU; this single-thread stack can't afford a biological-size expansion in evolution. Removing the compute tax unblocks every experiment above.

10. Reproducibility

- Deterministic by construction (seeded LCG-free Xoshiro; no HashMap iteration order).
- Frozen default identity: ``world_survival 20260924 77 60`` $\Rightarrow$
``WL2 RESULT beats=60 died_at=none final_energy=40.0 food_touches=0 novel_flags=0``.
- Test rig: ``KCPROBE`/`SW_PROBE`/`MB_PROBE`/`NEST_PROBE`` (env-gated, stderr) + registered
``SL2_*`/`WL2_*` knobs; workspace suite 198/0`.
- Workflow discipline captured in the ``anima-experiment-protocol`` managed skill.

Appendix A — Reproduction specification

Everything needed to recreate the experiments, with exact equations and registered constants. All mechanisms are **env-gated**; with no env set, the default run and its frozen output are byte-identical to the recorded baseline.

A.1 World & body dynamics

- Start energy **100.0**; ``SL2_PROG`` overrides to **PROG_LIFE 4000.0** (life-span mode).
- Per beat: ``energy -= 1.0`` (plus ``explore`` metabolic cost ``METABOLIC_SPEED × |vel|``); on food contact within **6.0** units, ``energy += 30.0``; death exactly at ``energy == 0``.
- ``SL2_STASIS`` (stable food): food does **not** teleport on touch. Re-entry guard — a touch counts

only if the body left the **6.0** eat radius since the last counted touch; tank caps at the life-start energy (``min(energy, tank_cap)``; ``tank_cap`` re-read after any PROG override).

A.2 Network construction (``build_net``)

Defaults: ``connectivity 0.038, w_init 0.2, amplitude 52.0, adaptation_tau_ms 200.0, adaptation_gain 0.05, inhibition_gain 0.0, slow_state_tau_ms 5000.0, slow_state_beta 0.0046875, latch_enable true, theta_rel_mean 1.0, theta_rel_sd 0, u_plateau_rel_sd 0, tau_het_rel_sd 0, phi_rel 0.5, eta_rel 0.0, output_inhibition_gain 0.0, d58_reflex K`` (K = the 8-neuron reflex/familiarity band that follows the outputs; see ``const K: usize = 8``). Env overrides: ``WL2_OINH``  $\rightarrow$  ``output_inhibition_gain 0.15``; ``SL2_POOL``  $\rightarrow$  internal pool size (default 40); ``SL2_MB``  $\rightarrow$  pool **256**, ``connectivity 0.10``; ``SL2_IINH``  $\rightarrow$  ``internal_inhibition_gain``; ``SL2_THETA``  $\rightarrow$  ``theta_rel_mean``; ``SL2_AMPL``  $\rightarrow$  ``amplitude``; ``SL2_HET``  $\rightarrow$  ``theta/u_plateau/tau_het_sd`` (per-neuron lognormal, applied after wiring). Wiring is seeded-via-``Xoshiro`` with no ``HashMap`` iteration order, so identical args+env $\Rightarrow$ identical network.

A.3 Fitness per scoring family (evolution ``evolve_run``)

D-38 death gate: a dead organism scores **0** unconditionally. Alive:

- **Plain-survival**: `fit = final_energy + 3·touches` (TOUCH_BONUS = 3.0)`
- **Density** (`SL2_DENSITY``): `+ 20·touches² / beats``
- **ARS** (`SL2_ARS``, subsumes density): `+ 4·persistence``, where
`persistence = ΣT exp(− gap_k / 20)`` over consecutive touches (beats since previous touch)
- **NESTFIT** (`SL2_NESTFIT``, adds to any alive base): `+ 40 · nest_returns``
`(`nest_returns` = count of false→true re-entries into the nest radius — see A.5)`

A.4 Behavioral drives (env-gated)

- **Hunger**: `hunger = 1 − energy / start_energy``.
- **SL2_HMEXP** (hunger-modulated exploration): motor-noise gain
`noise_scale = 0.4 + 0.6·hunger`` (fed ⇒ 0.4× baseline noise, hungry ⇒ 1×). Applied to all three explore-noise sites.
- **SL2_RSTDPP** (three-factor): on food touch set `dopamine = 1.0``; per beat `dopamine × = 0.6``;
STDP gain `= min(1 + 3·dopamine, 4)``.
- **SL2_NOVELTY** (MCC selection): behavior characterization `bc = [x_bin, z_bin, alive]`` where
`x_bin = clamp((px+50)/10, 0..9)``, same for z; novelty of an agent = `Σ_archive (diff(a,bc))/3 /`
`max(|archive|,1)``; selection rank key `(alive, novelty, fit)`` descending; archive cap **50** (evict oldest).

A.5 Home / rest (`SL2_NEST`` + `SL2_NESTFIT``)

- Nest = the non-food scene primitive nearest the spawn (deterministic).
- **Stamina** (capability constraint): starts 100; away from the nest,
`stamina − = 0.4 × |vel|`` per beat; within `NEST_RADIUS = 8.0`` of the nest, `stamina + = 60`` (cap 100).
Motor rates scale by `min(1, stamina / 25)`` — exhaustion immobilizes.
- `nest_returns`` increments on each false→true entry into the 8.0 nest radius.

A.6 Mushroom-body locus (`SL2_MB``)

Sparse expansion (256 KC, ~6 inputs/KC) + reward-gated input plasticity at the KC band: per touch, for every KC with eligibility `e > 0``, its input synapses get `Δw = 0.03 · e · 8`` (clamp 1.0). Measured saturation (A.7) currently makes `e > 0`` for all 256 KCs, so the bump is non-selective.

A.7 De-saturation measurements (the four-lever campaign)

`KCPROBE`` (env-gated) reports distinct internal-band spikers per sampled beat (`KCPROBE EVERY`` sets the beat interval). Results: pairwise competition 0.5–4.0 ⇒ 256/256; threshold 2–8× ⇒ 256/256; amplitude 52/20/8 ⇒ 256/256, amplitude 3 ⇒ 0/256 (a cliff); heterogeneity 0.2–1.0 ⇒ 256/256 alone, and at `amplitude=4, het=1.0`` the band is temporally bistable: beat 0 = 2/256, beat 20 = 3/256, beat 30+ = 256/256 (a sharp phase transition).

A.8 Verification battery and frozen identity

1. Build: `cargo build --release -p anima-world --bin world_survival``. 2. Default identity (no env): `./target/release/world_survival 20260924 77 60`` must print exactly `WL2 RESULT beats=60 died_at=none final_energy=40.0 food_touches=0 novel_flags=0``. 3. Determinism: run any gated config twice, `diff`` empty. 4. Suite: `cargo test --workspace`` → 198 passed / 0 failed. 5. Probe output is on **stderr** — capture with `2>&1 >/dev/null | grep``, not `2>/dev/null``.

A.9 Provenance notes

The **36-touch** (STASIS) and **2230-beat** (`SL2_PROG``) figures are recorded-lesson values

(session-learn log); logs for those long runs were overwritten by later runs, so they carry recorded provenance rather than a surviving artifact file. Every other cited number in this paper was verified directly against a surviving run log.

A.10 Genetic algorithm structure (`evolve\_run`)

- Population $N = 4$ per generation; seeded `net = build_net(net_inputs(smell), derive_seed64(esec,0,i))`.`

- Per-generation curriculum ramp: `gen_start = min(8 + 3·g, 40)`.`

- Selection: deterministic sort by fitness (or `(alive, novelty, fit)` under `SL2\_NOVELTY`);

death gate (dead $\Rightarrow$ fit 0) holds.

- **Elite carry**: `order[0]` carried unchanged (weights copied via `breed_inner`, mutate=false``);

after the same seed is best for **ELITE_STALL_GENS** = 5, the elite is carried **with** mutation (Fix B, anti-stall).

- Offspring: best $\rightarrow$ 2, second $\rightarrow$ 1; `breed_inner(parent, derive_seed64(esec,g+1,slot), mutate=true)``

copies parent weights by `(pre,post)`` and mutates each with  $p = 0.10$ , amplitude  $0.10$  (`W_MUT_P` / W_MUT_AMP``), then re-adds parent-born V2 synapses while under the growth cap.

- `SL2_NOTARGET`` suppresses the early TARGET break (`best > 40``); `--generalize`` is champion mode (probes the elite per gen on `BEATS_LONG`` natural food, stops at `ULTIMATE_BAR``).

- Beats per life: `BEATS_EVOLVE = 100`` (evolve+explore), `BEATS_LONG`` for champion probes.

A.11 Env-gate $\rightarrow$ exact-run-command table

Core (documentation only, not all commands run daily):

- Default / identity gate: `./target/release/world_survival 20260924 77 60``

- Stability + settle + home reward (full stack):

```
SL2_STASIS=1 SL2_HMEXP=1 SL2_NEST=1 SL2_NESTFIT=1 WL2_OINH=1 SL2_POOL=128
SL2_NOTARGET=1 ./target/release/world_survival --evolve --explore --smell --until 15`
```

- Per-mechanism observability runs use the same binary with a single gate added; probes

(`KCPROBE`, MB_PROBE`, SW_PROBE`, NEST_PROBE``) are env-gated and print to **stderr** — capture `2>&1 >/dev/null | grep <probe>`.`

- Seeds used: `20260924`` and `7`` (both `EVOLVE_SEEDS``).

11. Selected References

- FlyWire adult Drosophila connectome and its simulation in embodied bodies (NeuroMechFly, Nat.

Methods 2024; whole-brain embodied fly sims; Fly-connectomic graph models) — evidence that small spiking networks drive 3D behavior and that large-scale sims run real-time on GPU.

- Mushroom-body associative learning (sparse random expansion + dopamine-gated input plasticity;

trace conditioning; biased PN-KC connectivity) — the biological scheme ANIMA's `SL2_MB`` was drawn from, whose sparse-coding premise the saturation wall currently denies.

- Novelty search / minimal-criterion coevolution (Lehman & Stanley; MCC) — the selection-level

change that observably steers but does not escape the ceiling.

All results in this paper are reproducible from the repository; every claim is grounded in a measured run recorded in the session-learn log. Unverified causal attributions are explicitly marked.